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Symmetry, averageness, and feature size in the facial attractiveness of women

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Abstract

Female facial attractiveness was investigated by comparing the ratings made by male judges with the metric characteristics of female faces. Three kinds of facial characteristics were considered: facial symmetry, averageness, and size of individual features. The results suggested that female face attractiveness is greater when the face is symmetrical, is close to the average, and has certain features (e.g., large eyes, prominent cheekbones, thick lips, thin eyebrows, and a small nose and chin). Nevertheless, the detrimental effect of asymmetry appears to result solely from the fact that an asymmetrical face is a face that deviates from the norm. In addition, a factor analysis indicated that averageness best accounts for female attractiveness, but certain specific features can also be enhancing.
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1. Introduction

Attractiveness is an important element of social life. A large body of research in social psychology has shown that attractive persons enjoy many advantages that unattractive persons do not have (for reviews, see Berscheid & Walster, 1974; Bull & Rumsey, 1988; for a meta-analysis, see Eagly, Ashmore, Makhijani, & Longo, 1991). This well-known stereotype, described by social psychologists as the *What is beautiful is good* prototype (Dion, Berscheid, & Walster, 1972), mainly applies to the formation of first impressions, but it can also extend into less superficial interactions (e.g., Frieze, Olson, & Russell, 1991).

Until recently, beauty was considered to lie in *the eyes of the beholder* (for a historical background, see Bruce & Young, 1998). In that sense, a person's physical appearance is not the main aspect of his/her attractiveness. It is determined instead by the beholder, who probably takes personality into account in addition to physical appearance. Above all, it is believed that each beholder has personal likings that may differ from those of others. Thus, a face may be attractive for some people but unattractive for others. However, some recent studies have reported substantial agreement, not only within the sexes but also across sexes, ethnic groups, and ages (e.g., Cunningham, Roberts, Barbee, Druen, & Wu, 1995; Langlois, Ritter, Roggman, & Vaughn, 1991; Langlois & Roggman, 1990). Moreover, some facial characteristics have been shown to be factors of attractiveness, including closeness to the average (e.g., Langlois & Roggman, 1990), symmetry (e.g., Thornhill & Gangestad, 1993), and the physical characteristics of individual facial features (e.g., Cunningham, 1986). From an evolutionary standpoint, the fact that humans share views about which features are attractive suggests that there are species-typical psychological adaptations that have evolved because of a preference for healthy and fertile mates (Symons, 1979; Thornhill & Gangestad, 1999).

Working on the closeness-to-average factor, (Langlois & Roggman, 1990; see also Grammer & Thornhill, 1994) created composite computer faces by averaging photographs of women or men. The composite faces were assumed to be increasingly closer to the average as the number of faces that went into generating them rose. The authors found that by increasing the number of faces averaged into the composite, they also increased their attractiveness ratings. This observation was found for both female and male faces. Later, this finding was extended to other cultures (Langlois, Roggman, & Musselman, 1994; Pollard, 1995). The caricature versus anti-caricature technique (see Rhodes, Brennan, & Carey, 1987) has also been used to study the relationship between averageness and attractiveness. A caricatured face is a face in which the distinctive features in a set of faces are exaggerated by increasing the distance between the features and their average positions. Decreasing this distance makes the resulting face closer to the average, thereby creating anti-caricatured faces. In studies by Rhodes and Tremewan (1996) and Deffenbacher, Vetter, Johanson, and O'Toole (1998), anti-caricatured faces were deemed more attractive than normal faces, themselves being rated as more attractive than caricatured faces.

Thus, it appears that moving a face closer to the average increases its attractiveness. This is consistent with a previous report in which typical faces (more difficult to recognize) were judged more attractive than distinctive ones (more easily recognized; Light, Hollander, & Kayra-Stuart, 1981). Symons (1979) suggested that averageness is associated with a good phenotypic condition because of a stabilizing effect of natural selection on facial features. More recently, Thornhill and Gangestad (1993, 1999) suggested that a preference for average traits may have evolved because averageness denotes genetic heterozygosity, which could signal an outbreed mate or provide genetic diversity in defense against parasites.

On the other hand, some studies also indicate that, even if an average face is attractive, the most attractive faces are not close to the average. Shepherd and Ellis (1973), for example, did not obtain the linear relationship that Light et al. (1981) did between facial attractiveness and memorability, the latter dimension being linked to distinctiveness. They reported a U-shaped relationship. This suggests that distinctive faces comprise both the least and the most attractive faces. Perrett, May, and Yoshikawa (1994) provided support for this possibility using the caricature technique. They showed that the average face among a set of sixty women was less attractive than the average face of the fifteen most attractive women in the set. Moreover, caricaturing of the latter composite face increased its attractiveness. In conclusion, an average face is attractive, but the most attractive faces are not average ones.

A face's symmetry has also been reported as a factor of its attractiveness. Some studies indicate that symmetrical female and male faces are seen as more attractive than asymmetrical ones, and that having a symmetrical face is an advantage in sexual competition (Gangestad, Thornhill, & Yeo, 1994; Grammer & Thornhill, 1994; Mealey, Bridgstock, & Townsend, 1999; Thornhill & Gangestad, 1993). In the same way, many studies have suggested that facial symmetry reflects the person's phenotypic and genetic condition (e.g., Waynforth, 1998) as well as his/her psychological state (e.g., Shackelford & Larsen, 1997). So, like closeness to the average, symmetry has been proposed as a "certificate of health" in a mating perspective.

However, the role of symmetry has been questioned. Some authors have concluded that symmetry has little influence on attractiveness (Farkas, 1994; Jones & Hill, 1993). Others have found no impact at all (Langlois et al., 1994; Shackelford & Larsen, 1997). Langlois et al. (1994) suggested that only highly asymmetrical faces are unattractive; and finally, in three studies, asymmetrical normal faces were found to be more attractive than symmetrical composite versions of those same faces (Kowner, 1996; Samuels, Butterworth, Roberts, Graupner, & Hole, 1994; Swaddle & Cuthill, 1995). However, the techniques used in the latter studies have been criticized (Rhodes, Proffitt, Grady, & Sumich, 1998; Thornhill & Gangestad, 1999). Kowner (1996) and Samuels et al. (1994) used composite faces created by aligning a hemi-face with its mirror image. This technique may create abnormalities in facial feature size, such as a large nose. Swaddle and Cuthill (1995) averaged normal full faces with their mirror images, but they did not control facial expression. Using the same technique as Swaddle and Cuthill (1995) but with neutral expressions only, Rhodes et al. (1998) found that chimerical symmetrical faces were more attractive

than their original normal faces. As a whole, then, it seems that symmetrical faces are more attractive than asymmetrical ones.

The third approach used to study facial attractiveness consists in measuring certain facial features or manipulating feature size. For female faces, Cunningham (1986) (see also Keating, 1985) showed that women are more attractive when they have big eyes, a small nose, thin jaws, prominent cheekbones, a large mouth, high eyebrows, and so forth. These observations were later confirmed by Cunningham et al. (1995), who extended them to faces and judges from other cultures. Cunningham and his collaborators (1986, 1995) proposed the "Multiple Fitness Model," which states that attractiveness depends on multiple dimensions, with each feature representing a different aspect of mate value. They distinguished five dimensions, each related to specific facial features. Three of the dimensions are biological and age-related (neonatal, sexual maturity, and senescence) and two are influenced by more personal and social variables (facial expression and grooming). Neonatal features are thought to suggest desirable qualities of youthful vivaciousness, open-mindedness, and agreeableness. Sexual maturity features indicate postpubescent status and may suggest strength, dominance, status, and competency. They may also indicate an effective immune system, healthy resistance to parasites, and proper functional adaptation to the environment. Senescence features may convey non-competitive dominance. Expressive features suggest happiness, congeniality, non-threatening interest, social approachability, excitement, and arousal, whereas grooming features indicate successful adaptation, group membership, and status (see Cunningham et al., 1995). The attractiveness of a face depends on the harmonious presence of these different aspects (except for the features of senescence). Cunningham et al. (1995) proposed that an attractive face has neonatal features in the center of the face and sexual maturity features at the periphery. In contrast to the symmetry or closeness-to-average hypotheses, this model explains how two faces that seem attractive can nonetheless be distinctive.

According to the literature, an attractive woman has a symmetrical, close-to-average face with large eyes, a small nose, and so forth. However, at this time, no authors appear to have jointly studied the role of each of these characteristics. Rhodes, Sumich, and Byatt (1999) showed that an average face is not deemed attractive solely on the basis of its symmetry, but they did not include feature size in their studies. Many questions remain unanswered. Notably, what is the role of each of these characteristics? Is a woman more attractive with a symmetrical face, an average face, or large eyes? Some ensuing questions also emerge: What is the relationship between facial symmetry and facial averageness? One can assume that an average face is also symmetrical, but a symmetrical face is not necessarily close to the average. For example, one may have perfectly symmetrical eyes that are lower in the face than in other faces. Another question concerns the averageness and size of some features. It seems contradictory to find that a face is more attractive with both close-to-average features (e.g., Langlois & Roggman, 1990) and some larger (or thinner) features (e.g., Cunningham, 1986). For example, is it more important to have large eyes or average eyes? The answer is probably that it all depends on the feature in question.

Having close-to-average features implies having no unattractive features, which would make the face attractive. In this case, deviations from the average would enhance or decrease attractiveness according to (i) which feature is the deviating one, and (ii) the direction of the deviation. This suggestion is in line with the studies above which showed that close-to-average faces are attractive, but the most attractive faces are not close to average (e.g., Perrett et al., 1994).

In order to study the respective roles of symmetry, averageness, and feature size in the facial attractiveness of women, we parametrized these characteristics on a set of female faces. Attractiveness ratings of these faces were also collected. A first goal of the experiment was to confirm that symmetry, averageness, or feature size influence facial attractiveness. So, correlations were computed between attractiveness ratings and measures of symmetry, averageness, and feature size factors. According to the litterature, each of these factors should be correlated with ratings. A second goal was to study the weight of each of these factors. Partial correlations and factor analyses were computed to test if they had an independent influence on attractiveness ratings and if they had the same weight.

2. Experiment

2.1. Method

2.1.1. Participants

Eight males between the ages of 21 and 27 (mean = 24 years) volunteered to participate. They did not know the women whose faces were used in the experiment.

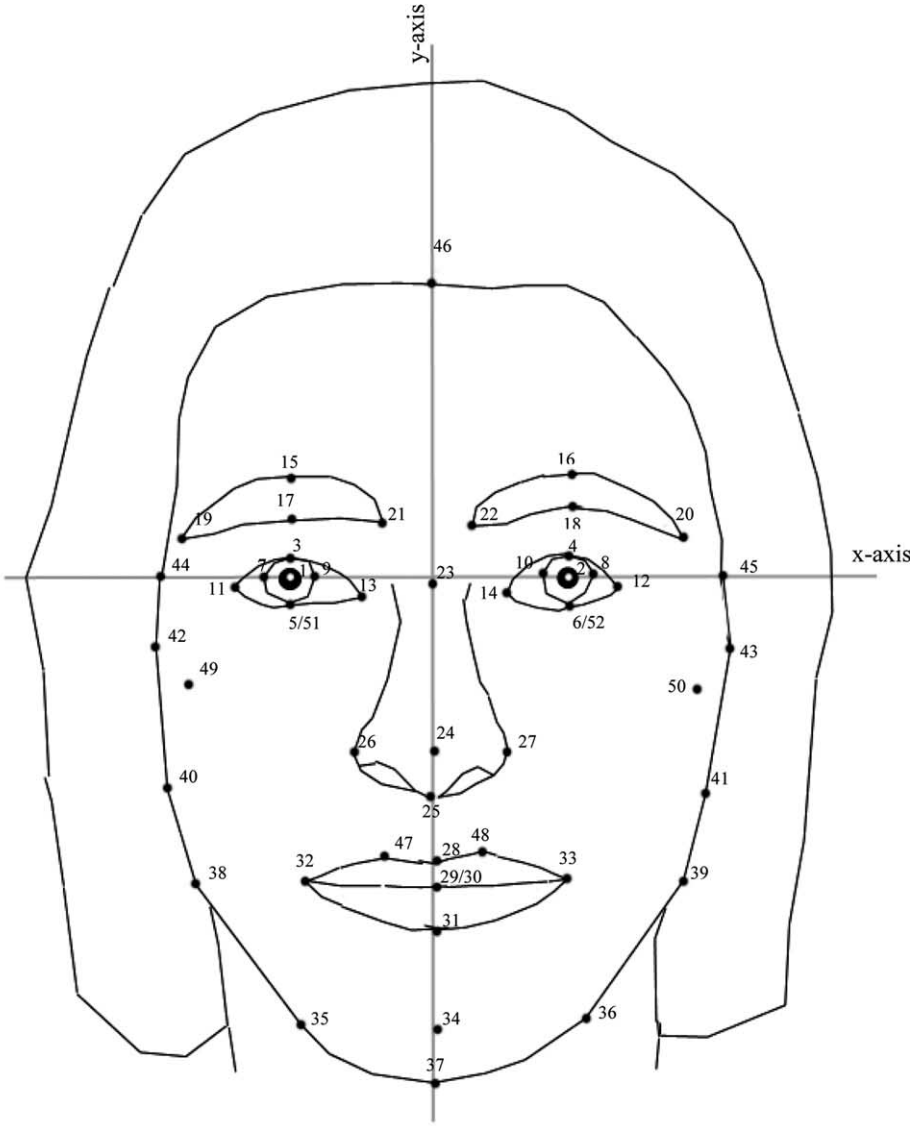
2.1.2. Material

Front-view photographs of 62 young adult women were used. Some of them came from an Internet face database (Stirling University face database). The others were photographed with an Ion2 or Kodak digital camera. The photographs were 20×20 cm. Face height was 12.5 cm from the top of the forehead to the tip of the chin. The center of the picture was halfway between the eyes. All information other than facial features was removed (see Fig. 1 for an example).

2.1.3. Procedure

The study had two stages: (i) evaluation of female face attractiveness by male participants, and (ii) face measurement.

2.1.3.1. Evaluation of female face attractiveness. The face of one woman was presented on the left side of the screen and the face of another woman was presented on the right side. The participants were told to press a key on the left or right side of the keyboard, depending on which woman they found the most attractive. Every face was presented with every other face. So, the number of times a woman was chosen as the most attractive indicated her attractiveness: the most attractive face could



1/2: left/right eye center; 3/4: left/right top of eye; 5/6: left/right bottom of iris; 7/8: outer edge of left/right iris; 9/10: inner edge of left/right iris; 11/12: outer edge of left/right eye; 13/14: inner edge of left/right eye; 15/16: upper edge of left/right eyebrow; 17/18: lower edge of left/right eyebrow; 19/20: outer edge of left/right eyebrow; 21/22: inner edge of left/right eyebrow; 23: bridge of nose; 24: tip of nose; 25: bottom of nose; 26/27: left/right edges of nose; 28/29: upper/lower edge of upper lip; 30/31: upper/lower edge of lower lip; 32/33: left/right mouth e dge; 34: tip of chin; 35/36: left/right face edge at chin level; 37: bottom of chin; 38/39: left/right face edge at mouth level; 40/41: left/right face edge at nose level; 42/43: left/right face edge at widest point; 44/45: left/right face edge at eye cent er; 46: top of face at hairline; 47/48: left/right tip of lip; 49/50: left/right tip of cheekbone; 51/52: left/right bottom of eye (if different from 5/6); 53: number of pixels of hair. Same *x*-coordinate: 1 -3-5-15-17-51; 2-4-6-16-18-52; 28-29-30-31. Same *y*-coordinate: 1-7-9; 2-8-10; 34-35-36; mean of 29/30 -38-39; 25-40-41; mean of 1/2 -44-45.

Fig. 1. Example of parametrized facial points.

have a score as high as 61 (i.e., she was judged more attractive than the other 61 women) and the least attractive woman could have a score as low as 0 (i.e., she was judged less attractive than the other 61 women). Each participant saw 1891 pairs presented in random order. The faces remained on the screen until the participant responded. There was no time limit. The left or right position of the two faces was alternated across participants.

2.1.3.2. Face measurements. Facial metrics. For each face, we measured the (x, y) coordinates of 53 points (see Fig. 1). The coordinates were determined by the authors using Adobe Photoshop software (version 5). Each measure was made twice, two months apart. The purpose of this double measurement procedure was to compensate for possible variability in measurement accuracy. Some points are easy to measure accurately, such as the iris, where the edges are clearly outlined and the contrast is sharp. Some other points are less clear-cut and can lead to greater measurement differences (e.g., nose or thin lips). Thus, measuring each point twice lessened the effect of such variations.

The (x, y) coordinates of each point were positioned in a Euclidean two-dimensional orthonormal space. The vertical axis was the face's vertical axis of symmetry. To determine this axis, a line was drawn on each photograph in order to cut the face into two hemi-faces. This approach is similar to the technique used to generate symmetrical composite faces from the right or left hemi-face (Kowner, 1996; Samuels et al., 1994). In the present study, the equation of this line was determined from the coordinates of two points on the line, one at the top and the other at the bottom of the photograph. Then, the x-coordinate of each point was converted into the distance from that line. Thus, the x-coordinate used in this study corresponded to the point's distance from the vertical axis of symmetry. The points to the left of that axis had a negative value, whereas the points to the right had a positive value. To obtain the measures for each point, the equation was determined twice. The first measures were located relative to the equation determined the first time, and the second measures were located relative to the equation determined the second time. The horizontal axis was the axis perpendicular to the vertical axis at the height of the eye center. When the eyes were not at the same height, the horizontal axis crossed the vertical axis at the mean eye height. The y-coordinate of each point corresponded to the point's distance from the horizontal axis. The points above that axis had a positive value whereas the points below it had a negative value.

Standardization. In order to normalize the measures, each distance was standardized as the ratio of the distance from four kinds of norms. The norms were (i) iris width, (ii) face length (i.e., distance from hairline to tip of chin), (iii) face width at eye height (distance between outer edges of face at eye center height), and (iv) area of face (length multiplied by width). Iris width was the most stable measure across faces. It was used in estimating the real size of other features. Another advantage is that the iris is round, which makes its width approximately equal to its height. Face length and width were the norms used by Cunningham et al. (Cunningham, 1986; Cunningham et al., 1995). They used face length for vertical measures (e.g.,

nose height) and face width for horizontal measures. We standardized all measures with these norms since this could be informative for both vertical and horizontal measures. For vertical measures, for example, a feature (e.g., mouth) may be perceived as wider when the face is shorter. The area of the face is particularly interesting for measures of the area of a given feature (e.g., area of the nose). A feature seems bigger when the face is small.

Measures of facial symmetry. In order to obtain an indicator of symmetry, we calculated a symmetry value for each point. For double points (i.e., ones with a "mirror" point on the other side of the face, such as eye center), horizontal or y-symmetry and vertical or x-symmetry were computed. Horizontal symmetry was equal to the absolute value of the difference between the y-coordinates of the two points (e.g., y-coordinate of the left eye center minus y-coordinate of the right eye center). Vertical symmetry was equal to the absolute value of the sum of the x-coordinates of the two points (e.g., x-coordinate of the left eye center plus x-coordinate of the right eye center). For single points (i.e., points located on the face's vertical axis of symmetry, such as the tip of the nose), only vertical or x-symmetry was computable. It was equal to the absolute value of the x-coordinate. The symmetry value of each point for each face was converted into a z-score by subtracting the mean for all faces from this value and dividing the result by the standard deviation for all faces. So, a z-score equal to 0 indicated that the point had an average amount of symmetry compared to the population of female faces. For a given point, a negative z-score indicated a high degree of symmetry whereas a positive z-score indicated a high degree of asymmetry. Overall facial symmetry as well as overall vertical and horizontal facial symmetry were computed by averaging the vertical and/or horizontal symmetry z-scores of all points. Similarly, the overall symmetry of single and double points was computed. For points with the same x- (or y-) coordinate (for example, the top and bottom of the eyes and eyebrows have the same x-coordinate as the eye's center), only the results for the reference point (in this example, the eye center) are reported. Similarly, the symmetry of these points was considered only once, when mean symmetry was computed.

Measures of facial averageness. The mean x- and y-coordinates and the standard deviation were calculated for each point. Then, for each point of each face, its distance from the mean position was calculated by subtracting its x-coordinate from the mean x-coordinate and its y-coordinate from the mean y-coordinate. The distance from the mean was then computed using the Pythagorean theorem. The x and y distances were considered as two sides of a right triangle. This gave us three indicators of the distance from the average: x distance, y distance, and overall distance. As for symmetry, all distances were converted into z-scores by subtracting the mean distance for all faces from the individual distance, and dividing the result by the standard deviation among all faces. So, the closer the faces were to the average, the greater the negative z-scores. For each face, taking the mean of the averageness scores of all points gave its overall averageness. Similarly, overall x- and y-averageness were computed by taking the mean of the x- and y-averageness scores of all points. As for the analysis of the symmetry effect, only the value of the reference point was considered when different points were aligned.

Measures of feature size. The size of various facial features was determined from the standardized point measures. The size measures were basically the same as some of the ones used by Cunningham and collaborators (Cunningham, 1986; Cunningham et al., 1995). The distance between two features, as well as the width, height, length, and thickness of the features were all determined by computing the distance between two points. For double features (e.g., eyes, eyebrows, cheekbones), the size reported was the mean of the two elements. Most distances were computed using the Pythagorean theorem, where the difference between the x-coordinates was taken as the length of one side of a right triangle and the difference between the y-coordinates was taken as the length of the other side. The distance between the two points was the length of the longest side of the right triangle. Some distances were computed by considering the absolute value of the difference between the x- or y-coordinates. This was the case when the two points had the same y- or x-coordinate. The area of a given feature was calculated by multiplying its width by its height.

The sizes computed were eye width (distance 11–13 and 12–14), eye height (distance 3–51 and 4–52), area of eye (eye width × eye height), horizontal eye separation (distance 1–2), eyebrow thickness (distance 15–17 and 16–18), eyebrow height (distance 1–17 and 2–18), eyebrow length (distance 17–19 + 17–21 and 18–20 + 18–22), nose width (distance 26–27), nose length (distance 23–25), area of nose (nose width × nose length), mouth height (distance 28–31), mouth width (distance 32–33), area of mouth (mouth height × mouth width), upper lip thickness (distance 28–29), lower lip thickness (distance 30–31), cheekbone prominence (width of face at cheekbones divided by width of face at mouth), cheekbone height (y-coordinates of 42 and 43), chin length (distance 31–37), chin width (distance 35–36), area of chin (chin length x chin width), width of face at eyes (distance 44–45), width of face at cheekbones (distance between x-value of 42 and 43), width of face at nose (distance 40–41), width of face at mouth (distance 38–39), and amount of hair (number of pixels in hair region).

Design. In order to investigate the role of each facial characteristic in facial attractiveness ratings, we calculated the Bravais–Pearson coefficient of correlation between the attractiveness score and each facial measure. We also analyzed the extent of agreement among judges. In every analysis, the number of items considered was the number of faces ($N = 62$). To study the weight of each characteristic in the final evaluation, partial correlations as well as factor analyses were computed.

2.2. Results

2.2.1. Agreement among judges

Agreement among judges was assessed by calculating the correlation between the attractiveness scores of each judge and the attractiveness score given by all other judges, either considered individually or taken together (by averaging their responses). All coefficients were significant (see Table 1). Thus, the judges as a whole agreed about the attractiveness of the female faces. This observation is in line with previous studies (e.g., Cunningham, 1986; Cunningham et al., 1995).

Table 1
Agreement between each judge and each other judge, considered either individually or together (Bravais–Pearson *r*)

	Judge 1	Judge 2	Judge 3	Judge 4	Judge 5	Judge 6	Judge 7	Judge 8
Judge 1	1							
Judge 2	0.72[a]	1						
Judge 3	0.75[a]	0.71[a]	1					
Judge 4	0.63[a]	0.71[a]	0.66[a]	1				
Judge 5	0.53[a]	0.67[a]	0.62[a]	0.46[b]	1			
Judge 6	0.63[a]	0.55[a]	0.55[a]	0.44[b]	0.62[a]	1		
Judge 7	0.49[c]	0.55[a]	0.53[a]	0.56[a]	0.39[d]	0.37[d]	1	
Judge 8	0.64[a]	0.68[a]	0.67[a]	0.63[a]	0.35[d]	0.49[c]	0.48[c]	1
Total	0.81[a]	0.84[a]	0.82[a]	0.73[a]	0.65[a]	0.65[a]	0.59[a]	0.70[a]

[a] $p < 0.00001$.
[b] $p < 0.001$.
[c] $p < 0.0001$.
[d] $p < 0.01$.

2.2.2. Facial attractiveness and facial symmetry

The coefficients obtained for overall symmetry, mean vertical symmetry, mean vertical and horizontal symmetry of double points, and mean vertical symmetry of single points are reported in Table 2. Correlations for individual points are also reported when of interest.

The correlation between attractiveness and overall symmetry was significant and negative: the more asymmetrical a woman's face, the less attractive it was judged to be. However, the analysis of the different types of asymmetry showed that only double point asymmetry had a significantly detrimental effect on attractiveness. Neither overall horizontal symmetry nor overall single point symmetry was significantly correlated with attractiveness. For double points, only vertical asymmetry was significantly correlated with attractiveness; horizontal asymmetry was not. Furthermore, an individual analysis of each point indicated that the vertical symmetry of certain double points was significantly and negatively correlated with attractiveness. This was true for eye center and the outer/inner edges of the iris, as well as for the edges of the face at the height of the eyes, cheekbones, nose, and mouth. Thus, the harmful effect of double point vertical asymmetry was mainly based on two facial characteristics: the eye region and facial boundaries. Moreover, while the overall horizontal symmetry of double points was not significantly correlated with attractiveness, some of these points were correlated, mainly in the eye region (i.e., eye centers and edges). No other points were significantly correlated with attractiveness ratings.

In summary, a woman was seen as less attractive when her face was asymmetrical. However, only two main sources of asymmetry really had an effect: the vertical and horizontal asymmetry of the eye region and the vertical asymmetry of the facial edges. In other words, a female face is less attractive when its eyes are not at the same height on the face (i.e., horizontal asymmetry), when the eyes are not at the same

Table 2
Correlation between women's facial attractiveness and facial symmetry (Bravais–Pearson r), according to four norms

Facial point	Iris	Face length	Face width	Face area
Vertical symmetry	***−0.29***[a]	***−0.30***[b]	***−0.28***[a]	***−0.29***[a]
Horizontal symmetry	***−0.17***	***−0.17***	***−0.15***	***−0.15***
Double point symmetry	***−0.35***[b]	***−0.37***[b]	***−0.35***[b]	***−0.35***[b]
Double point vertical symmetry	***−0.30***[a]	***−0.31***[a]	***−0.30***[a]	***−0.30***[a]
Eye center	−0.24[c]	−0.24[c]	−0.24[c]	−0.24[c]
Iris outer edge	−0.25[a]	−0.26[a]	−0.26[a]	−0.26[a]
Iris inner edge	−0.27[a]	−0.28[a]	−0.27[a]	−0.27[a]
Face edge at eye level	−0.32[a]	−0.32[a]	−0.32[a]	−0.32[a]
Face edge at cheekbone level	−0.31[a]	−0.31[a]	−0.31[a]	−0.31[a]
Face edge at nose level	−0.32[a]	−0.32[a]	−0.32[a]	−0.32[a]
Face edge at mouth level	−0.30[a]	−0.30[a]	−0.30[a]	−0.30[a]
Face edge at chin level	−0.21[d]	−0.21[d]	−0.21[d]	−0.21[d]
Double point horizontal symmetry	***−0.17***	***−0.17***	***−0.15***	***−0.15***
Eye center	−0.28[c]	−0.29[c]	−0.27[c]	−0.28[c]
Iris upper edge	−0.23[d]	−0.24[d]	−0.22[d]	−0.23[d]
Eye outer edge	−0.26[c]	−0.25[c]	−0.24[c]	−0.24[c]
Eye inner edge	−0.22[d]	−0.22[d]	−0.22[d]	−0.22[d]
Eye upper edge	−0.25[d]	−0.25[c]	−0.23[d]	−0.24[d]
Single point symmetry	***−0.17***	***−0.17***	***−0.15***	***−0.15***
Overall Symmetry	***−0.34***[b]	***−0.35***[b]	***−0.34***[b]	***−0.34***[b]

[a] $p < 0.02$.
[b] $p < 0.01$.
[c] $p < 0.05$.
[d] $p < 0.10$.

distance from the face's center (vertical asymmetry), or when one half of the face is larger than the other.

2.2.3. Facial attractiveness and facial averageness

Correlation coefficients for overall averageness, mean x-averageness (i.e., distance from the x-coordinate mean), and mean y-averageness (i.e., distance from the y-coordinate mean) are reported in Table 3. Correlations for individual points are also reported whenever significant or useful.

Overall averageness was significantly and negatively correlated with attractiveness. Thus, female faces were seen as more attractive when closer to the average. This was true for overall averageness as well as for x- and y-averageness. However, a look at the averageness of individual points indicated that averageness was more important for some facial features than for others. This was true for the edges of the face and eyes, the mouth, and the area of the nose, where averageness was significantly and negatively correlated with attractiveness. The eyes and mouth had a detrimental effect on attractiveness when they moved away from the average along the y-axis (the correlation was not significant along the x-axis). For the nose, a y-coordinate that moved away was more detrimental than an x-coordinate, but this effect was amplified if the $x$moved away too, and overall averageness on such points was the most

Table 3
Correlation between women's facial attractiveness and facial averageness (Bravais–Pearson r), according to four norms

Facial point	Iris	Face length	Face width	Face area
Left face edge at eye level (x-averageness and overall averageness)	−0.13	−0.05	−0.29[a]	−0.10
Right face edge at eye level (x-averageness and overall averageness)	−0.26[a]	−0.29[a]	−0.24[b]	−0.29[a]
Left and right face edges at nose level (y-averageness)	−0.18	−0.27[a]	−0.08	−0.15
Left face edge at nose level (x-averageness)	−0.11	−0.11	−0.14	−0.11
Left face edge at nose level (overall averageness)	−0.24[b]	−0.32[c]	−0.18	−0.20
Right face edge at nose level (x-averageness)	−0.30[c]	−0.29[a]	−0.33[d]	−0.31[c]
Right face edge at nose level (overall averageness)	−0.33[d]	−0.37[d]	−0.27[a]	−0.31[c]
Left and right face edges at chin level (y-averageness)	−0.08	−0.19	−0.21	−0.21[b]
Left face edge at chin level (x-averageness)	−0.12	−0.15	−0.12	−0.16
Left face edge at chin level (overall averageness)	−0.12	−0.23[b]	−0.19	−0.25[a]
Right face edge at chin level (x-averageness)	−0.35[d]	−0.40[d]	−0.37[d]	−0.37[d]
Right face edge at chin level (overall averageness)	−0.31[c]	−0.40[d]	−0.35[d]	−0.39[d]
Left eye center (x-averageness)	−0.11	−0.07	−0.12	−0.10
Left eye center (y-averageness)	−0.33[d]	−0.34[d]	−0.33[d]	−0.33[d]
Left eye center (overall averageness)	−0.31[c]	−0.30[c]	−0.34[d]	−0.33[d]
Right eye center (x-averageness)	−0.12	−0.02	−0.10	−0.17
Right eye center (y-averageness)	−0.33[d]	−0.34[d]	−0.33[d]	−0.33[d]
Right eye center (overall averageness)	−0.34[d]	−0.31[c]	−0.36[d]	−0.40[d]
Left nose edge (x-averageness)	−0.06	−0.07	−0.05	−0.03
Left nose edge (y-averageness)	−0.19	−0.32[c]	−0.15	−0.19
Left nose edge (overall averageness)	−0.17	−0.26[a]	−0.10	−0.13
Right nose edge (x-averageness)	−0.09	−0.13	−0.16	−0.15
Right nose edge (y-averageness)	−0.19	−0.33[d]	−0.12	−0.23[b]
Right nose edge (overall averageness)	−0.25[b]	−0.33[d]	−0.24[b]	−0.30[c]
Left mouth corner (y-averageness)	−0.01	−0.19	−0.07	−0.22[b]
Right mouth corner (y-averageness)	−0.04	−0.25[a]	0.05	−0.15
Upper edge of upper lip (y-averageness)	−0.23[b]	−0.24[b]	−0.18	−0.23[b]
Lower edge of upper lip (y-averageness)	−0.30[c]	−0.26[a]	−0.21	−0.23[b]
Upper edge of lower lip (y-averageness)	−0.24[b]	−0.26[a]	−0.18	−0.19
Lower edge of lower lip (y-averageness)	−0.11	−0.20	−0.20	−0.19
Left tip of lip (y-averageness)	−0.26[a]	−0.30[c]	−0.19	−0.25[a]
Right tip of lip (y-averageness)	−0.24[b]	−0.32[c]	−0.20	−0.26[a]
Amount of hair	0.22[b]	0.22[b]	0.24[b]	0.24[b]
X-AVERAGENESS	*−0.38*[d]	*−0.33*[d]	*−0.35*[d]	*−0.42*[d]
Y-AVERAGENESS	*−0.33*[c]	*−0.41*[d]	*−0.28*[a]	*−0.35*[d]
OVERALL AVERAGENESS	**−0.46**[d]	**−0.48**[d]	**−0.43**[d]	**−0.52**[d]

[a] $p < 0.05$.
[b] $p < 0.10$.
[c] $p < 0.02$.
[d] $p < 0.01$.

significant factor. By contrast, the edges of the face mainly had a detrimental effect when they moved away from the average along the x-axis. All significant correlations

were negative, i.e., the effect of moving away from the average was always detrimental. There was only a tendency ($p < 0.10$) towards a positive correlation for the amount of hair.

To summarize the effects of facial averageness, attractiveness benefitted from averageness either at an overall level, or for many individual facial features (e.g., eyes, nose, mouth, face shape). Moreover, for the individual features (eyes, mouth, nose), the detrimental effect of moving away from the average was observed mainly when the movement was along the y-axis. For most of these features, and mainly for the points that delineated them (e.g., mouth and nose edges), the present analysis could not tell us anything about what kind of deviation from the average really acted upon the attractiveness ratings. Two main explanations are possible: the feature was larger than the average or it was not in the same place on the face. Further analysis of feature size could point to the correct explanation. However, for points that indicated the position of a feature (e.g., eye center), the results indicated that it was the fact of being in a different spatial position from the average that decreased attractiveness.

2.2.4. Facial attractiveness and feature size

The size of certain facial features was computed in order to study their correlation with attractiveness. The results for each size measure are reported in Table 4.

Some features were significantly and positively correlated with attractiveness, namely, eye width, mouth and upper-lip thickness, and cheekbone prominence and height. Some other features were significantly but negatively correlated with attractiveness: brow thickness, and width and area of the nose and chin. The other features studied were not significantly correlated with attractiveness.

2.2.5. Symmetry and averageness in facial attractiveness

We calculated partial correlations in order to study the hypothesis that facial asymmetry has a detrimental effect simply because having an asymmetrical face amounts to having a non-average face, and vice versa. The correlation between attractiveness and (i) facial averageness, and (ii) facial symmetry were computed by keeping the other dimension constant. The results are presented in Table 5.

Symmetry and averageness were found to be significantly correlated with attractiveness (see previous findings). Moreover, symmetry and averageness were significantly and positively correlated with each other: moving away from perfect symmetry concomitantly produced moving away from the average (and reciprocally). Since the three variables co-varied in pairs, one can suspect that a correlation between any two of them simply results from the fact that each one co-varies with the third. Partial correlation analysis validated this assumption. Whereas the correlation between averageness and attractiveness was always significant when symmetry was kept constant, the correlation between symmetry and attractiveness was no longer significant when averageness was kept constant. In other words, the symmetry effect in attractiveness ratings resulted solely from the fact that asymmetrical faces tended to move away from the average. Note that the correlation between symmetry and averageness was always significant, even when attractiveness was held constant.

Table 4
Correlation between women's facial attractiveness and facial feature size (Bravais–Pearson *r*), according to four norms

Feature	Iris	Face length	Face width	Face area
Eye width	0.27[a]	0.28[a]	0.38[b]	0.32[c]
Eye height	−0.06	−0.06	−0.02	−0.03
Eye area	0.02	0.03	0.08	0.06
Horizontal eye separation	0.03	0.03	0.10	0.07
Eyebrow thickness	−0.29[a]	−0.30[c]	−0.27[a]	−0.28[a]
Eyebrow height	0.13	0.14	0.16	0.15
Eyebrow length	−0.02	−0.03	0.04	0.01
Nose width	−0.27[a]	−0.27[a]	−0.24[d]	−0.22[d]
Nose length	−0.19	−0.23[d]	−0.13	−0.18
Nose area	−0.32[c]	−0.35[b]	−0.28[a]	−0.28[a]
Mouth height	0.29[a]	0.28[a]	0.30[c]	0.29[a]
Mouth width	−0.09	−0.10	−0.01	−0.05
Mouth area	0.22[d]	0.21[d]	0.25[d]	0.22[d]
Upper lip width	0.33[b]	0.33[b]	0.34[b]	0.33[b]
Lower lip width	0.17	0.16	0.19	0.18
Cheekbone prominence	0.32[c]	0.32[c]	0.32[c]	0.32[c]
Cheekbone height	0.28[a]	0.29[a]	0.26[a]	0.27[a]
Chin length	−0.11	−0.13	−0.07	−0.10
Chin width	−0.36[b]	−0.37[b]	−0.36[b]	−0.36[b]
Chin area	−0.30[c]	−0.32[c]	−0.28[a]	−0.29[a]
Face width at eyes	−0.10	−0.13	0.00	−0.08
Face width at cheekbones	−0.14	−0.17	−0.16	−0.14
Face width at nose	−0.17	−0.18	−0.14	−0.14
Face width at mouth	−0.20	−0.22[d]	−0.18	−0.18
Amount of hair	0.14	0.14	0.15	0.15

[a] $p < 0.05$.
[b] $p < 0.01$.
[c] $p < 0.02$.
[d] $p < 0.10$.

Table 5
Single and partial correlations between attractiveness ratings, symmetry, and averageness

	Iris	Face length	Face width	Face area
Simple correlation				
Symmetry and attractiveness	−0.34[a]	−0.35[a]	−0.34[a]	−0.34[a]
Averageness and attractiveness	−0.46[a]	−0.48[a]	−0.43[a]	−0.52[a]
Symmetry and averageness	0.40[a]	0.60[a]	0.54[a]	0.49[a]
Partial correlation (third dimension constant)				
Symmetry and attractiveness	−0.19	−0.09	−0.14	−0.11
Averageness and attractiveness	−0.38[a]	−0.36[a]	−0.31[b]	−0.43[a]
Symmetry and averageness	0.29[c]	0.53[a]	0.46[a]	0.39[a]

[a] $p < 0.01$.
[b] $p < 0.02$.
[c] $p < 0.05$.

Table 6
Proportion of variance (R^2) explained by each facial characteristic

Facial characteristic	R^2									
Number of variables	1	2	3	4	5	6	7	8	9	10
Overall averageness	0.2494									
Chin width	0.1325	0.3084	0.3628	0.3760	0.4356	0.4995	0.5733	0.6287	0.6469	0.6492
Left eye center averageness	0.1293	0.2611	0.3186	0.3885	0.4641	0.5032	0.5794	0.6323	0.6477	0.6510
Right eye center averageness	0.1208	0.2660	0.3258	0.3843	0.4371	0.4912	0.5733	0.6304	0.6483	0.6505
Overall asymmetry	0.1183	0.2747	0.3264	0.3799	0.4375	0.4920	0.5794	0.6285	0.6461	0.6491
Upper lip thickness	0.1060	0.2898	0.3555	0.4353						
Face edge asymmetry at nose	0.1054	0.3106	0.3539	0.4000	0.4718	0.5303	0.6285			
Cheekbone prominence	0.1054	0.3107	0.3760							
Face edge asymmetry at eyes	0.1029	0.2916	0.3373	0.3911	0.4506	0.5127	0.3023	0.6310	0.6468	0.6493
Nose area	0.1005	0.2743	0.3263	0.4141	0.4911					
Face edge asymmetry at cheekbones	0.0969	0.3002	0.3425	0.3936	0.4619	0.5196	0.6225	0.6291	0.6482	0.6509
Face edge asymmetry at mouth	0.0909	0.3088	0.3576	0.3959	0.4672	0.5269	0.6235	0.6288	0.6461	0.6487
Double point vertical asymmetry	0.0892	0.2780	0.3289	0.3803	0.4394	0.4954	0.5921	0.6287	0.6462	0.6491
Chin area	0.0879	0.2970	0.3504	0.3761	0.4353	0.4929	0.5735	0.6292	0.6466	0.6488
Eyebrow thickness	0.0849	0.3113								
Mouth height	0.0814	0.2831	0.3568	0.4250	0.4450	0.4995	0.5791	0.6460		
Eye center horizontal asymmetry	0.0788	0.2574	0.3169	0.3821	0.4413	0.4916	0.5738	0.6285	0.6468	0.6492
Cheekbone height	0.0761	0.2937	0.3429	0.3968	0.4443	0.4912	0.5787	0.6293	0.6487	
Nose width	0.0732	0.2770	0.3303	0.3934	0.4865	0.4952	0.5735	0.6298	0.6462	0.6488
Eye width	0.0703	0.2916	0.3626	0.4137	0.4667	0.5733				
Eye center vertical asymmetry	0.0560	0.2623	0.3160	0.3763	0.4358	0.4923	0.5810	0.6285	0.6461	0.6487
Total						0.6712				

2.2.6. Respective roles of symmetry, averageness, and feature size: a factorial analysis

R^2 was calculated for all facial parameters found to be correlated with attractiveness in the previous analysis. This index was used to indicate the amount of variance in attractiveness that was explained by an individual parameter. The first R^2 column in Table 6 gives the results for all parameters considered individually (the measures were normalized with respect to iris width). They are listed in descending order of R^2. The other R^2 columns give the results when we added each variable to the previous most significant variable(s).

The facial parameter that best explained variations in the attractiveness ratings was overall averageness ($R^2 = 0.2494$). The second parameter was chin width ($R^2 = 0.1325$), which explained about half as much variance. The subsequent parameters ranged from 0.1293 (left eye center averageness) to 0.0560 (eye center vertical symmetry). When all parameters were pooled, R^2 was about 0.6712. Thus, overall averageness was the main factor that determined facial attractiveness. It accounted for nearly 25% of the variance in the attractiveness ratings. More than two-thirds of the variance was explained by the parameters defined in the present experiment.

In order to determine what parameters enhanced attractiveness the most when combined with averageness, we conducted a multiple regression analysis by adding each individual parameter to overall averageness (see Table 6). The optimal multiple regression analysis was the one where eyebrow thickness was associated with overall averageness ($R^2 = 0.3113$). The subsequent parameters that gave rise to an increase in the explained variance were (in descending order) cheekbone prominence ($R^2 = 0.3760$), upper lip thickness ($R^2 = 0.4353$), area of nose ($R^2 = 0.4911$), eye width ($R^2 = 0.5733$), symmetry of face edges at nose ($R^2 = 0.6285$), mouth height ($R^2 = 0.6460$), and cheekbone height ($R^2 = 0.6487$). All remaining parameters gave rise to small and roughly equivalent increases in the amount of explained variance.

2.3. Conclusion

In summary of the results observed in this study, all facial parameters reported in the literature were found to influence female attractiveness. A women was seen as more attractive when her face was symmetrical and close to the average, and when its features had certain characteristics like big eyes, a small nose, prominent and high cheekbones, a small chin, and a thick mouth and upper lip. However, the analysis of the weight of each of these characteristics indicated that the main factor of attractiveness was averageness. Moreover, the effect of facial symmetry in the attractiveness ratings appeared to result from an association between facial symmetry and averageness. It seems that an asymmetrical face is not unattractive merely because it is asymmetrical, but because the asymmetry implies moving away from the average. Finally, the fact that some facial features were not of average size but, on the contrary, were larger or smaller, sometimes increased attractiveness. This was true here when the cheekbones were more prominent and higher, the upper lip or mouth was thicker, the area of the nose was smaller, and the eyes were wider. Consequently, even if averageness was the most important facial characteristic in attractiveness, the most attractive female faces were more than just close to the average.

3. General discussion

One of the main conclusions of this study is that the positive impact of facial symmetry in attractiveness evaluations of female faces mainly results from the fact that a symmetrical face tends to be closer to the average than an asymmetrical face. This observation is not surprising if we consider that an average face is also symmetrical. There appear to be two different ways of deviating from the average. First, some facial features may move in different ways, say, one eye may be higher than the average, while the other is lower or in the same position. In this case, the face becomes asymmetrical as well as being different from the average. A second way to deviate occurs when a feature is different from the average but is still symmetrical. This may be the case, for example, when the two eyes are higher in the face than the average, but are at the same height on the face and are equally spaced on either side of the middle. Both of these kinds of deviation have a negative influence on attractiveness. Thus, there may be deviations from the average that do not modify facial symmetry, but it is not possible to alter facial symmetry without moving away from the average. Consequently, variations in symmetry are a subset of variations in averageness. As underlined above, this hypothesis rests on the postulate that an average face is symmetrical. To test this postulate, we computed the overall symmetry of a face whose coordinates were the mean coordinates of the 62 faces used in the present study, and obtained a value of −1.12. The mean symmetry of all faces was 0 (SD = 0.45) and the overall symmetry of the most symmetrical face in our set was −0.98. Thus, the average face was indeed the most symmetrical face, which tends to confirm the idea that an average face is also symmetrical.

The second conclusion is that facial averageness is the characteristic that contributes the most to facial attractiveness in women. However, some other characteristics, mainly some aspects of individual features, made a substantial contribution to the attractiveness of an average face. This observation is in line with studies showing that an average face is attractive, but the most attractive faces are not average (e.g., Perrett et al., 1994). Being close to the average for a face appears to be the basis of attractiveness. In other words, the main characteristic that prevents a face from being unattractive is the absence of local deviations from the average. Once this condition is met, the face may be more or less attractive, depending on whether the local deviations have a positive or negative effect. Thus, in conclusion, the attractiveness of a female face depends on (i) the averageness of the majority of its features, and (ii) deviation from the average of some local features. Moreover, the implications of the deviation depend on (i) the feature that deviates, and (ii) the direction of the deviation. Finally, variations around the average have neither a symmetrical nor a linear effect on attractiveness.

In an evolutionary and mating perspective, the finding reported in the present study may be accounted for by two hypotheses: that averageness is associated with a good phenotypic condition (Symons, 1979) or heterozygosity (Thornhill & Gangestad, 1993), and that an enhanced size of certain features may indicate biological and social value for reproduction (Cunningham, 1986; Cunningham et al., 1995). An average, symmetrical face may act as a health certificate. Deviations from this

average—notably, asymmetrical deviations—may indicate environmental perturbations during development, genetic problems, or psychological distress (Shackelford & Larsen, 1997; Thornhill & Gangestad, 1993, 1999). Nonetheless, some deviations may also be indicative of biological or social attributes that are useful in mate selection (Cunningham et al., 1995). For women in particular, a high estrogen-to-testosterone ratio in puberty results in enlargement of the lips and the upper cheek caused by fat deposits. The present study, like many others (e.g., Cunningham, 1986; Cunningham et al., 1995; Keating, 1985) showed that these facial characteristics enhance the facial attractiveness of women. Such features may be used as sex-hormone markers (Thornhill & Gangestad, 1999), i.e., indicators of a postpubescent status and fertility. Other features may also have a biological or social value, suggesting youthful vivaciousness, strength, dominance, non-threatening interest, excitement, and so forth, which further enhances facial attractiveness. A harmonious distribution of these characteristics is probably preferable, as suggested by Cunningham et al. (1995), but it seems that they may be organized around an average face. The number of facial characteristics that affect attractiveness explains how two apparently distinctive faces may both be attractive. A face may be attractive because it has strong sex-hormone markers, even if it is a little asymmetrical or somewhat removed from the average. Another face with fewer markers can be just as attractive by having more symmetrical or closer-to-average features. Similarly, some variations across judges are observed in accordance with the weight each one grants to the different characteristics.

A multidimensional model of facial attractiveness, as proposed by Cunningham and his collaborators (1986, 1995), is the best way to account for attractiveness. Such a model must take four main principles into account. (1) Not all dimensions have the same weight, with the most important one being closeness to the average. One may consider that the origin of these dimensions (i.e., their intersection point) corresponds to an average and symmetrical face. (2) Some dimensions are dependent upon others. Symmetry, for example, is a particular case of averageness. (3) A linear variation on these dimensions does not necessarily give rise to linear or Gaussian changes in attractiveness. One can observe non-linear or asymmetrical patterns of attractiveness that are contingent upon the dimension and the direction of the variation. (4) Each dimension is more or less associated with biological and/or social values that inform the observer about the physiological and psychological health of a perspective mate.

This study allowed us to show that different characteristics of a face play a role in its attractiveness, and to different degrees. Nonetheless, some points remain unresolved. In particular, the relationship between symmetry and averageness is not clear. We now know that moving away from the average decreases attractiveness even if the face is still symmetrical (Rhodes et al., 1999), and that moving away from symmetry amounts to moving away from the average. However, we do not know whether the impact of the deviation is the same for symmetrical and asymmetrical shifts. In the same way, the present study suggests that there are various kinds of relationships between attractiveness and feature size, and that they differ across features. However, features were not manipulated individually here, so, other aspects of

the face were not controlled. Furthermore, the effect of the size of some features might not have appeared in this study because of variations in the others. Future research could look into how attractiveness evolves as the different dimensions of a face are varied while other dimensions are held constant.

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